P2P CHAT APPLICATION

Software Requirement Specifications

Version 1.0

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification provides a complete description of all the functions and constraints of the p2p chat application. This application will be implemented based on C/C++, Java and other open source software.

1.2 Scope

This R&D project arms to develop a free p2p chat application that everyone can use

1.3 Definitions, Acronyms and Abbreviations

Term	Definitions, Acronyms and Abbreviations

1.4 References

P2p chat app GUI

2. OVERAL DESCRIPTION

2.1 Product Perspective

This Java#19 project is a new, self-contained product. The product we about to develop will be a stand-alone application in that it will not be part of any system.

The software will be implemented as PC-based application using Java,C/C++ language and open source software. It will be comprised of technologies as J2SE, Swing, Spring, Log4J, Junit, Java Speech AP, mySQL

2.2 Product Features

Following main features will be made available for the software. The features' detail will be described at "System Features" section.

- Client
- Account
- Chatting
- Main Server
- Stun Server
- Relay Server

2.3 User Documents

As with any good application, documentation and help facilities will exist for P2P chat application. The following documents will be made available to users:

• Installation guide

• User guide/document

2.4 Assumptions and Dependencies

This document has been worked out based on the following assumptions:

2.4.1 Technical Assumptions

2.4.2 Managerial Assumptions

2.4.3 Dependencies

2.5 Constraints

Project is to be developed / delivered as Open Source (spring, mySQL, speex ,boost library…).

3. USE CASE MODEL

The Use Case Model defines the behavior of the system (functional specification) from the users' perspective. It includes the Actors that interact with the system (such as customers and the ABC web store employees), and the Use Cases that describe these interactions.

3.1 The Actors

An Actor is someone outside the system that communicates directly with the system.

Admin: Admin is a root authority person, with technical role he is responsible for setting up and maintaining the servers

User: normal user who uses p2p chat application client to chat with each other

#	Main features	Admin	User
1	Client		x
2	Account		x
3	Chatting		x
4	Main Server	x	
5	Stun Server	x	
6	Relay Server	x	

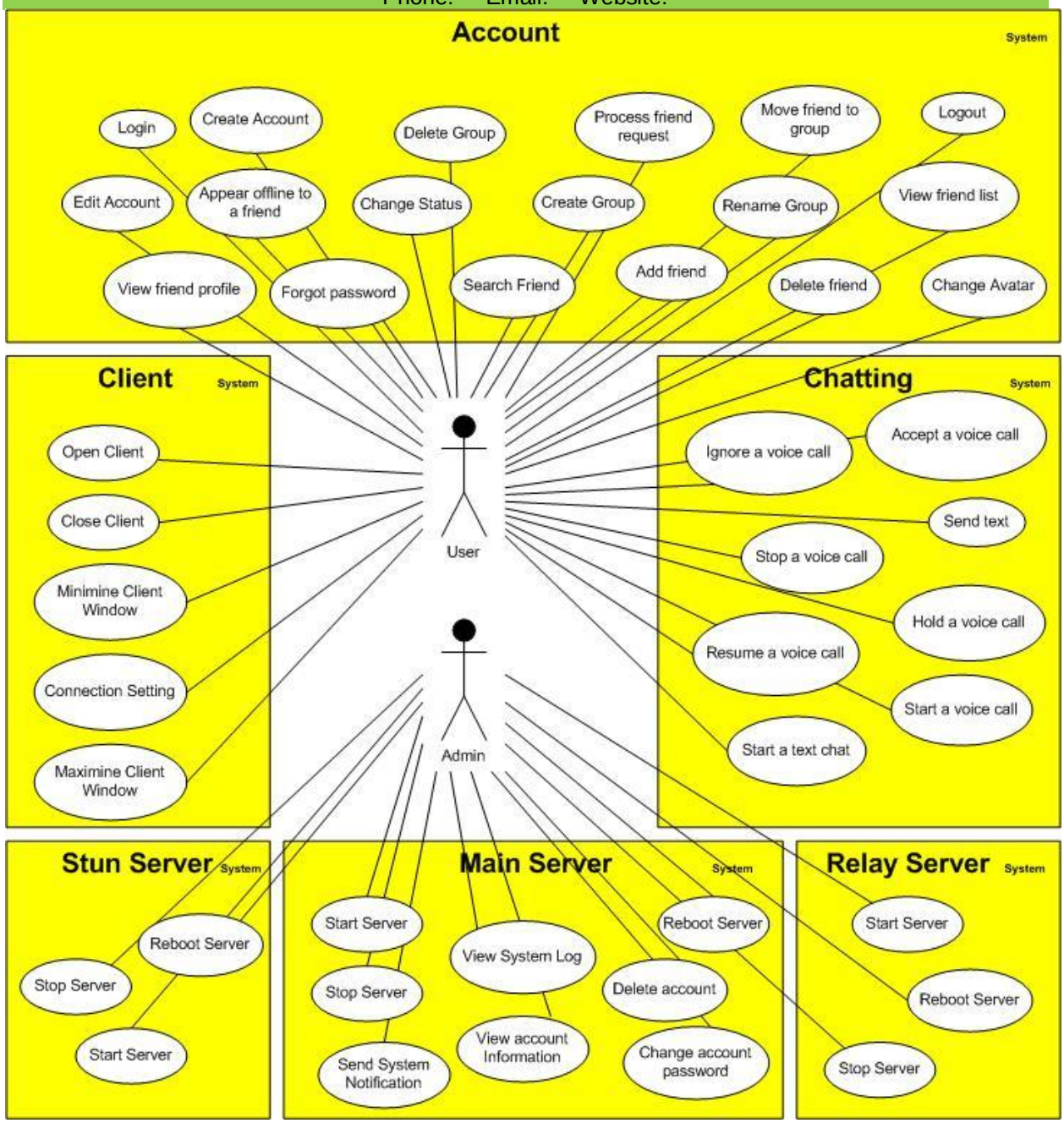
3.2 The use cases

A Use Case is a description of a complete flow of events that results in something of value to an Actor

System Features	Main Use Cases	Use Case ID
Client		
	Open Client	Cli-001
	Close Client	Cli-002
	Minimize Client Window	Cli-003
	Maximize Client Window	Cli-004
	Connection Setting	Cli-005
Account		
	Create Account	Acc-001
	Login	Acc-002
	Edit Account	Acc-003
	Logout	Acc-004
	Forgot password	Acc-005
	Change Avatar	Acc-006
	Change Status	Acc-007
	Appear offline to a friend	Acc-008

	View friend list	Acc-009
	View friend profile	Acc-010
	Search Friend	Acc-011
	Add friend	Acc-012
	Delete friend	Acc-013
	Process friend request	Acc-014
	Create Group	Acc-015
	Delete Group	Acc-016
	Rename Group	Acc-017
	Move friend to group	Acc-018
Chatting		
	Start a text chat	Cha-001
	Send text	Cha-002
	Start a voice call	Cha-003
	Accept a voice call	Cha-004
	Ignore a voice call	Cha-005
	Stop a voice call	Cha-006
	Hold a voice call	Cha-007
	Resume a voice call	Cha-008
Main Server		
	Start Server	*Adm-001*
	Stop Server	*Adm-002*
	Reboot Server	*Adm-003*
	View System Log	*Adm-004*
	Change account password	*Adm-005*
	View account Information	*Adm-006*
	Delete account	*Adm-007*
	Send System Notification	*Adm-008*
Stun Server		
	Start Server	Stu-001
	Stop Server	Stu-002
	Reboot Server	Stu-003
Relay Server		
	Start Server	Rel-001
	Stop Server	Rel-002
	Reboot Server	Rel-003

3.3 Use cases diagram



4. SYSTEM FEATURES

4.1 Client

Use Case ID	Cli-001
Name	Change Connection Setting
Goal	Modify connection setting
Actors	User
Pre-conditions	User has not yet login
Post-conditions	

Main Flow	1. User click on "Connection Setting" 2. Show "Connection Setting" dialog 3. User change connection setting 4. User press apply button 5. User close the dialog	
Exceptions		
Open Issues	Wrong Connection Setting	

4.2 Account

Use Case ID	Acc-001	
Name	Create Account	
Goal	Want to sign up an account with server	
Actors	User	
Pre-conditions	N/A	
Post-conditions	User will receive a notification popup and confirmation email	
Main Flow	1. Register an account 3. Enter account information: Name, username, email, password, retype-password, … 7. Check Email 8. Confirm registration	2. Display Registration Windows 4. Check Account information 5. Add user to database 6. Send confirmation email 9. Send welcome email
Exceptions	4A incorrect input information 4B email already used 4C username already used 8A customer don't confirm registration	
Open Issues	N/A	

Use Case ID	Acc-002	
Name	Login	
Goal	User use client to login to system	
Actors	User	
Pre-conditions	User already has an account, Client already Open	
Post-conditions	User login to system successful	
Main Flow	1. Open login window 2. Input username + password and press OK	3. Check username and password in

	5. Show Login Successful windows 6. Show offline message popup	database 4. Send friend list, offline message to client
Exceptions	3A username or password incorrect, show notification	
Open Issues	N/A	

Use Case ID	Acc-003	
Name	Edit Account	
Goal	Change User account information	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	Update user account information	
Main Flow	1. Open "Edit Account" feature 3. Edit/update account information and press OK	2. Show account setting window 4. Update information to database 5. Send update information to user
Exceptions	4A User's information isn't validate, show notification	
Open Issues	N/A	

Use Case ID	Acc-004	
Name	Logout	
Goal	User logout from system	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	User logout from system, client show login window	
Main Flow	1. Press logout button on client. 3. Show login window	2. Remove client from online list
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-005	
Name	Forgot Password	
Goal	Send password in email to user's email	
Actors	User	
Pre-conditions	User has not yet login	
Post-conditions	N/A	
Main Flow	1. Press "Forgot Password"	

	2. Show "Forgot Password" dialog 3. Type user's email and press OK 6. Check email and get password 7. Use new password to login	4. Send password in email
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-006	
Name	Change Avatar	
Goal	Change User's avatar	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	User's avatar is changed	
Main Flow	1. Press on avatar 2. Show "My Display Image" Windows 3. Choose an image to change avatar	4. Update information to database 5. Send update avatar to user and user's friend
Exceptions	4a. If User want another avatar, press "Browse" button to browse more images and press OK to upload new image to server 4aa Image size is to big, show notification	
Open Issues	N/A	

Use Case ID	Acc-007	
Name	Change Status	
Goal	User's status is changed	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Press on status or status frame 2. Show list all available status and status frame 3. Choose a status or type status text. Press OK	4. Update new status to user and user's friend
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-008	
Name	View friend profile	
Goal	Show a friend profile dialog	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Right click on a friend in his list 2. Press "view profile"	3. Show friend profile dialog
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-009	
Name	Search Friend	
Goal	Show searching result dialog	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Type a name on search text field	2. Show searching result
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-010	
Name	Add Friend	
Goal	Add new friend to friend list	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Press add friend button 2. Type a username(User2) to add and press OK	4. Search username in database 5. Send request to User2 6. Update information to database 7. Send update friend list to clients(User1 and User2)
Exceptions	4a Username doesn't have in database, show notification to User1 4b User2 isn't login to system, show request dialog when User2 login 5a User2 accept friend request 5b User2 ignore friend request	
Open Issues	N/A	

Use Case ID	Acc-011	
Name	Delete friend	
Goal	Delete a friend from friend list	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Select a friend 2. Right click on friend 4. Choose delete friend from context menu and press OK	3. Show context menu 5. Update information to database 6. Send update friend list to client
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-012	
Name	Create Group	
Goal	Show a new group in friend list	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Choose “Create new group” from context menu 2. Show “Create group” dialog, type group name and press OK	3. Update information to database 4. Send update friend list to client
Exceptions	3A Group name has already exist, show notification.	
Open Issues	N/A	

Use Case ID	Acc-013	
Name	Delete group	
Goal	Delete a group from friend list	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Right click on a group 2. Choose “delete group” from context menu and press OK	3. Update information to database 4. Send update friend list to client
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-014	
Name	Rename group	
Goal	Rename a group from friend list	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	N/A	
Main Flow	1. Right click on a group 2. Choose “rename group” from context menu 4. Type new group name and press OK	3. Show “rename group” dialog 5. Update information to database 6. Send update friend list to client
Exceptions	4A Group name has already exist, show notification.	
Open Issues	N/A	

Use Case ID	Acc-015	
Name	Move a friend to a group	
Goal	Friend that has been chosen display in a new group	
Actors	User	
Pre-conditions	User has already login	
Post-conditions		
Main Flow	1. Right click on a friend in friend list. 2. Choose “move to group” in context menu 4. Choose a group from group list and press OK	3. Show a group list. 5. Update new information to database 6. Send update friend list to client
Exceptions	N/A	
Open Issues	N/A	

Use Case ID	Acc-016	
Name	Change Password	
Goal	Change user’s password	
Actors	User	
Pre-conditions	User has already login	
Post-conditions	Send an email to user	
Main Flow	1. Press “Change Password”	2. Show “Change Password” dialog

	3. Type old password, new password, new password confirm and press OK	4. Change user's password in database
Exceptions	3A New password and new password confirm does not match, show notification	
Open Issues	N/A	

4.3 Chatting

User Case ID	Cha-001	
Name	Start a text chat	
Goal	Display a chat window	
Actors	User	
Pre-conditions	Connection has established	
Post-conditions	N/A	
Main Flow	1. Start a text chat.	2. Display text chat frame.
Exception	N/A	
Open Issues	N/A	

User Case ID	Cha-002	
Name	Send text	
Goal	Send contents of text chat	
Actors	User	
Pre-conditions	Enabled text chat window	
Post-conditions	Display contents chat on text frame chat	
Main Flow	1. Send text	2. Display contents on text frame chat.
Exception	N/A	
Open Issues	N/A	

User Case ID	Cha-003	
Name	Start a voice call	
Goal	Send voice chat request	
Actors	User	
Pre-conditions	Connection has established	
Post-conditions	Appear voice chat request on friend's computer	
Main Flow	1. Start a voice call.	2. Waiting for acceptance from another customers
Exception	1A. If micro is not found, show error message 2A. If the user waiting over 30 seconds, show the dialog notification	
Open Issues	N/A	

User Case ID	Cha-004	
Name	Accept a voice call	
Goal	Accept a voice call from another user	
Actors	User	
Pre-conditions	Has a voice call request from another customers	
Post-conditions	Successful connection notification appears	
Main Flow	1. Accept a voice call.	2. Display voice call frame.
Exception	N/A	
Open Issues	N/A	

User Case ID	Cha-005	
Name	Ignore a voice call	
Goal	Ignore a voice call from another customers	
Actors	User	
Pre-conditions	Has a voice call request from another customers	
Post-conditions	Unsuccessful connection notification appears	
Main Flow	1. Ignore a voice call.	2. Disable voice call frame.
Exception	N/A	
Open Issues	N/A	

User Case ID	Cha-006	
Name	Stop a voice call	
Goal	Stop a voice call	
Actors	User	
Pre-conditions	Voice call established	
Post-conditions	Disable voice call frame	
Main Flow	1. Stop a voice call.	2. Disable voice call frame
Exception	N/A	
Open Issues	N/A	

User Case ID	Cha-007	
Name	Hold a voice call	
Goal	Hold a voice call	
Actors	User	
Pre-conditions	Voice call established	
Post-conditions	Resume button appears on user's chat window Hold connection notification appears on friend's user text frame chat	
Main Flow	1. Hold a voice call.	2. Pauses voice call
Exception	N/A	
Open Issues	N/A	

User Case ID	Cha-008	
Name	Resume a voice call	
Goal	Resume a voice call from another customers	
Actors	User	
Pre-conditions	A voice call was held.	
Post-conditions	Voice call frame is resumed Resume connection notification appears on user text frame chat and friend's user text frame chat	
Main Flow	1. Resume a voice call.	2. Resume voice call.
Exception	N/A	
Open Issues	N/A	

4.4 Main server

User Case ID	Adm-001	
Name	Start Server	
Goal	Want to start server.	
Actors	Administrator	
Pre-conditions	Server hasn't started yet.	
Post-conditions	Listen requests to clients	
Main Flow	1.Open Linux terminal 2. Enter command line like Linux services to start server.	3. Show message to announce that server started.
Exception	1A. if server is running then ,show message prompt that server has been already running 1B.If command line is not correct or suited, show error message and asks to input again.	
Open Issues	N/A	

User Case ID	Adm-002
Name	Stop Server
Goal	Want to stop server.
Actors	Administrator
Pre-conditions	Server is running
Post-conditions	N/A

Main Flow	1. Open Linux terminal 2. Enter command line like Linux services to stop server.	3. Show message to announce that server stopped.
Exception	2A.If command line is not correct or suited, show error message and asks to input again.	
Open Issues	N/A	

User Case ID	Adm-003	
Name	Reboot Server	
Goal	Want to reboot server.	
Actors	Administrator	
Pre-conditions	Server is servicing.	
Post-conditions	Server will service back.	
Main Flow	1. Open Linux Terminal 2. Enter command line like Linux services to reboot server, with option is maximum time which system take to spend.	3. Show message to announce that server will reboot.
Exception	2A.If command line is not correct or suited, show error message and ask to input again.	
Open Issues	Restarting, system can be power failure.	

User Case ID	Adm-004	
Name	View System Log	
Goal	Want to view system log.	
Actors	Administrator	
Pre-conditions	N/A	
Post-conditions	N/A	
Main Flow	1. Open Linux Terminal 2. Enter command line to view system log	3.Show sytem log
Exception	N/A	
Open Issues	N/A	

User Case ID	Adm-005
Name	Change account password
Goal	Want to change (an) account password(s).

Actors	Administrator	
Pre-conditions	This (These) user name(s) is(are) valid, and old password is correct, maybe security password.	
Post-conditions	Send to backup email a notification and confirmation mail.	
Main Flow	1. Open Linux Terminal 2. Enter command line to change account password 3. Choose and update information (password) of user(s) requesting to change password in database.	4. Display successful message, and show new information of user(s).
Exception	2A.If command line is not correct or suited, show error message and asks to input again.	
Open Issues	N/A	

User Case ID	Adm-006	
Name	View account Information	
Goal	Want to view (an) information of account(s).	
Actors	Administrator	
Pre-conditions	This (These) user name(s) is(are) valid.	
Post-conditions	N/A	
Main Flow	1. Open Linux Terminal 2. Enter command line to view account information	3. Show account information
Exception	2A.If command line is not correct or suited, show error message and ask to input again.	
Open Issues	N/A	

User Case ID	Adm-007
Name	Delete account
Goal	Want to delete account(s).
Actors	Administrator
Pre-conditions	This (These) user name(s) is(are) valid.
Post-conditions	Send to backup email an announced mail.

Main Flow	1. Open Linux Terminal 2.Enter command line to get account list 4.Enter command line to delete an account using account ID found in step 3	3.Show account list 5.Show result message
Exception	2A.If command line is not correct or suited, show error message and asks to input again. 4A.If command line is not correct or suited, show error message and asks to input again.	
Open Issues	N/A	

User Case ID	Adm-008	
Name	Send System Notification	
Goal	Want to announce to user(s).	
Actors	Administrator	
Pre-conditions	N/A	
Post-conditions	N/A	
Main Flow	1.Open Linux terminal 2.Enter command line to send message to all user	3. Display successful message.
Exception	N/A	
Open Issues	N/A	

4.5 Relay server

User Case ID	Rel-001	
Name	Start Relay Server	
Goal	Relay server start to work.	
Actors	Administrator.	
Pre-conditions	N/A	
Post-conditions	N/A	
Main Flow	1. Open Linux terminal 2. Enter command to start relay server	

		3.Show successful message
Exception	N/A	
Open Issues	N/A	

User Case ID	Rel-002	
Name	Stop Relay Server	
Goal	Server stop working	
Actors	Administrator.	
Pre-conditions	N/A	
Post-conditions	N/A	
Main Flow	1. Open Linux terminal 2. Enter command to stop relay server	3.Show successful message
Exception	N/A	
Open Issues	N/A	

User Case ID	Rel-003	
Name	Reboot Relay Server	
Goal	Server restarts.	
Actors	Administrator.	
Pre-conditions	N/A	
Post-conditions	N/A	
Main Flow	1. Open Linux terminal 2. Enter command to reboot server(option is time to reboot)	3. Show successful message
Exception	N/A	
Open Issues	N/A	

4.6 Stun server

User Case ID	Stu-001	
Name	Start Stun Server	
Goal	Stun Server starts working.	
Actors	Administrator	
Pre-conditions	N/A	
Post-conditions	N/A	
Main Flow	1. Open Linux terminal 2. Enter command to start Stun Server	3.Show successful message

Exception	N/A
Open Issues	N/A

User Case ID	Stu-002	
Name	Stop Stun Server	
Goal	Server stops working.	
Actors	Administrator.	
Pre-conditions	Server is working.	
Post-conditions	Server has stop	
Main Flow	1. Open Linux terminal 2. Enter command to stop Stun server	2. Show successful message
Exception	N/A	
Open Issues	N/A	

User Case ID	Stu-003	
Name	Reboot Stun Server	
Goal	Server restarts.	
Actors	Administrator.	
Pre-conditions	N/A	
Post-conditions	N/A	
Main Flow	1. Restart server. 2. Enter command to reboot Stun server(option is time to reboot)	3.Show successful message
Exception	N/A	
Open Issues	N/A	

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